

# Lab 3 - React Hooks

## Context Hook and Effect Hook

### Instructions

In this assignment you will add some React Hooks to the React application. Using your Lab 1, 2 for continuing this Lab.

### Objectives and Outcomes

In this assignment, you will continue to work with the React application that you have been developing in the Lab1 & Lab2. You will build a simple authentication system in React using:

- Context Hook for global state management.
- Effect Hook for side effects and localStorage synchronization.
- Custom Hook for reusable and cleaner code. (optional)

By the end of this lab, you will have a working login/logout mechanism where user state is persisted in localStorage and shared across multiple components.

### Lab Requirements

1. Create a toggle to change the dark and light theme for all elements at your React application.
2. For this lab, you must create the following components/modules with the specified names:

- AuthProvider – A provider component that supplies authentication state and functions to the rest of the app.
- AuthContext – A context object to be used for sharing authentication data.

- useAuth – A custom hook that allows components to easily access authentication state and functions. (optional)
- Dashboard or Navbar – A component that consumes authentication data and displays different UI depending on whether the user is logged in or not.
  - If logged in: “Welcome, {username}” and a Logout button.
  - If not logged in: “Log in as (ex: Aaron)” and a Login button.

**Additional Notes:**

- You may hardcode a sample username for login instead of creating a login form.
- No need to call an API or handle real authentication logic.
- The login action should simply set a fake user object in the global state.
- The logout action should clear this fake user from the global state.